

Project Title: Merging Datacentres into AI-Driven Manufacturing Through Intelligent Telemetry, In-Network Learning, and Self-Healing Infrastructure

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1.0 Introduction and Motivation

Manufacturing and process automation environments are undergoing a fundamental transformation as artificial intelligence workloads move closer to the factory floor. Modern industrial systems now resemble distributed datacentres, generating continuous, high-volume telemetry from sensors, robotic cells, edge servers, and AI inference pipelines. These systems demand ultra-low latency, high bandwidth, resilience, and energy efficiency, while operating under strict real-time constraints (Chinnaraju, 2024).

Existing manufacturing networks increasingly adopt high-performance interconnects and edge-cloud architectures to support AI-driven control, predictive maintenance, and autonomous optimisation. However, these infrastructures remain largely reactive, relying on static monitoring, rule-based thresholds, and control-plane-centric intelligence. This creates latency bottlenecks, limits responsiveness to failures, and restricts the scalability of AI-driven manufacturing systems (Polu, 2024; Zhang, 2025).

Recent advances in programmable data planes, in-band telemetry, and AI-driven optimisation provide an opportunity to re-architect manufacturing infrastructure as an intelligent, self-optimising system that merges datacentre-grade performance with industrial constraints (Aramide, 2025). This research proposes a novel framework that embeds intelligence directly into the network and telemetry layers of AI-driven manufacturing systems, enabling real-time decision-making, predictive failure analysis, and autonomous self-healing.

2.0 Background and Literature Review

2.1 AI Workloads and High-Performance Interconnects

The proliferation of terascale AI workloads, including distributed training, inference pipelines, and data-intensive analytics, has placed unprecedented pressure on network interconnects (Aramide, 2025). Technologies such as InfiniBand and Ultra Ethernet provide the physical capabilities required for high throughput and ultra-low latency. However, their full potential is constrained by insufficient telemetry density and limited AI-aware feedback mechanisms (Aramide, 2025; Manzoor, Raza and Domzal, 2024).

Aramide (2025) demonstrates that combining programmable data planes with fine-grained telemetry and AI-based optimisation can significantly improve latency, congestion management, and network health. Yet, these approaches have not been fully explored in industrial manufacturing contexts, where constraints on determinism, reliability, and energy efficiency are more stringent.

2.2 Programmable Data Planes and In-Network Intelligence

Programmable data plane technologies such as P4 and eBPF enable real-time customisation of packet processing, telemetry extraction, and flow control within the network fabric itself. Unlike fixed-function pipelines, these technologies allow observability and control to be embedded directly into the data path, reducing reliance on centralised controllers and improving responsiveness (Aramide, 2024).

Zhang (2025) extends this paradigm by embedding machine learning inference directly into the data plane, enabling autonomous, low-latency decisions without frequent control-plane interaction. This work highlights a critical limitation of traditional SDN architectures, where control-plane dependence introduces delays that are unacceptable for time-sensitive applications such as congestion mitigation and intrusion response.

2.3 AI-Driven Load Balancing and Edge Computing in Manufacturing

Edge computing plays a crucial role in AI-driven manufacturing by processing data close to its source. Zhang et al. (2025) demonstrate that deep reinforcement learning can effectively balance workloads across industrial edge

servers, reducing system latency and improving efficiency. However, these approaches largely treat the network as a passive substrate rather than an active participant in optimisation.

2.4 Predictive Failure Analysis and Self-Healing Systems

AI-driven predictive failure analysis has emerged as a key technique for improving resilience in cloud datacentres. Deep learning models such as LSTMs, Transformers, and GNNs have been shown to outperform traditional statistical and rule-based approaches in predicting failures from telemetry and system logs (Aramide, 2025; Polu, 2024).

The integration of AIOps, federated learning, and edge computing enables proactive failure detection while preserving privacy and reducing latency. Nevertheless, deploying these techniques in real-time, latency-critical environments remains challenging due to computational overhead and model adaptability (Gill et al., 2025; Polu, 2024).

Veeravalli (2026) argues that future AI systems must evolve into self-healing distributed networks, embedding intelligence across micro, meso, and macro layers of infrastructure. Despite promising conceptual frameworks, practical implementations face unresolved challenges in observability, complexity management, and validation.

3.0 Problem Statement and Research Gap

3.1 Problem Statement

AI-driven manufacturing environments are rapidly evolving into large-scale, distributed computing systems in which industrial sensors, robotic cells, vision systems, and control loops continuously generate high-volume, latency-sensitive data streams (Zhang, 2025). Unlike conventional cloud workloads, these systems impose strict real-time constraints, deterministic execution requirements, and energy budgets while operating under harsh physical and operational conditions. As a result, manufacturing floors increasingly resemble datacentres in scale and complexity yet remain fundamentally constrained by the need for ultra-low latency, high availability, and predictable behaviour (Polu, 2024).

Existing approaches based on Software-Defined Networking (SDN), Artificial Intelligence for IT Operations (AIOps), and edge orchestration frameworks are not sufficient to meet these requirements when applied in isolation or in their current forms. Traditional SDN architectures rely on centralised control planes that depend on frequent data-plane-to-controller communication. While effective for traffic engineering in enterprise and cloud networks, this design introduces non-negligible control latency that is incompatible with sub-millisecond reaction times required for congestion mitigation, failure response, and safety-critical manufacturing operations (Gill et al., 2025; Zhang, 2025). Moreover, SDN abstractions provide limited support for fine-grained telemetry and in-situ decision-making within high-speed interconnects such as InfiniBand and Ultra Ethernet, restricting their ability to respond dynamically to AI-driven traffic patterns (Zhang, 2025).

Similarly, AIOps platforms have achieved success in cloud environments by leveraging log aggregation, anomaly detection, and automated remediation. However, these systems are predominantly designed for offline or near-real-time analysis of historical data, with inference and decision-making occurring at centralised management layers. In industrial AI settings, where failures and congestion events can propagate within microseconds, such delayed, control-plane-centric responses are inadequate. Furthermore, AIOps solutions often assume elastic compute and network resources typical of hyperscale datacentres, an assumption that does not hold in energy-constrained manufacturing environments with fixed infrastructure and limited overprovisioning capacity (Polu, 2024).

Edge orchestration frameworks partially address latency by moving computation closer to data sources. However, current edge solutions focus primarily on application-level task placement and container scheduling, offering limited visibility into network-level dynamics and insufficient integration with high-performance interconnects. They typically lack mechanisms for continuous, high-resolution telemetry, in-network learning, and closed-loop optimisation across both compute and communication layers. As a result, edge orchestration alone cannot guarantee coordinated adaptation of networking, workload scheduling, and energy consumption under rapidly changing industrial AI workloads (Gill et al., 2025; Polu, 2024; Zhang et al., 2025).

Collectively, these limitations reveal a critical gap: *existing SDN, AIOps, and edge orchestration approaches were not designed for tightly coupled, real-time co-optimisation of networking, computation, and energy in industrial-grade AI systems*. They treat these dimensions as loosely connected subsystems rather than as an integrated, adaptive infrastructure. Consequently, they are unable to provide the resilience, responsiveness, and efficiency required for next-generation AI-driven manufacturing.

This research addresses this gap by investigating how programmable data planes, real-time telemetry, and distributed machine learning can be combined to enable low-latency, energy-aware workload orchestration and resilient networking directly within the manufacturing fabric. By embedding intelligence closer to the data path and enabling closed-loop adaptation across network and compute layers, the proposed approach aims to transcend the architectural limitations of existing solutions and establish a new foundation for industrial-grade AI infrastructure.

3.2 Research Gap

Synthesising the literature reveals a clear and underexplored gap at the intersection of AI-driven manufacturing, networked systems, and intelligent infrastructure:

1. Existing industrial and cloud systems rely heavily on **control-plane-centric intelligence**, introducing latency and limiting real-time responsiveness (Zhang, 2025).
2. Telemetry systems often lack the **resolution, integration, and AI-aware feedback loops** required for proactive optimisation in high-speed, distributed environments (Aramide, 2025).
3. Predictive failure analysis and self-healing mechanisms are typically implemented above the network layer, missing opportunities for **in-network intelligence and local autonomy** (Polu, 2024).
4. There is limited research on **merging datacentre-grade networking principles with manufacturing constraints**, particularly in the context of energy efficiency, determinism, and operational resilience.

4.0 Research Aim and Objectives

4.1 Aim

To design and evaluate an intelligent, self-optimising infrastructure for AI-driven manufacturing that merges datacentre-grade networking with industrial process automation through real-time telemetry, in-network learning, and autonomous resilience mechanisms.

4.2 Objectives

1. To design a high-resolution telemetry framework for AI-driven manufacturing networks using programmable data planes.
2. To embed machine learning inference within the network fabric to enable low-latency, local decision-making.
3. To develop AI-driven models for congestion prediction, load balancing, and predictive failure analysis using real-time telemetry.
4. To implement self-healing mechanisms that autonomously adapt network and system behaviour in response to failures and workload dynamics.
5. To evaluate the proposed framework in terms of latency, resilience, scalability, and energy efficiency in industrial AI scenarios.

5.0 Research Questions

1. How can programmable data planes and in-band telemetry enable real-time intelligence in AI-driven manufacturing networks?
2. How does embedding machine learning inference within the data plane reduce latency and improve resilience compared to control-plane-centric approaches?
3. What AI models are most effective for congestion prediction, load balancing, and predictive maintenance in distributed manufacturing systems?
4. How can self-healing mechanisms be designed to balance autonomy, efficiency, and reliability in industrial AI infrastructures?

6.0 Methodology and System Architecture

This section defines the end to end technical approach for designing, implementing, and evaluating a datacentre grade infrastructure stack adapted to AI driven manufacturing, combining high resolution telemetry, in network learning, predictive failure analysis, and energy aware orchestration.

6.1 System Overview and Architectural Layers

The proposed architecture is organised into five interacting layers:

1. Instrumentation layer (telemetry sources)

Switches, NICs, edge gateways, and compute nodes expose real time signals including queue depth, ECN marks, per flow RTT proxies, packet drops, link utilisation, CPU and GPU utilisation, power draw, and system logs. High granularity telemetry is necessary to support fast feedback loops for AI workloads in high-speed fabrics (Aramide, 2025; Veeravalli (2026).

2. Programmable data plane layer (in network telemetry + actuation)

Using programmable packet processing, the network embeds in band telemetry and supports local actuation primitives such as queue reconfiguration, rate limiting, path selection hints, and on switch feature extraction. Programmable data planes enable more flexible telemetry and control close to traffic, rather than relying only on controller side reasoning (Aramide, 2024; Aramide, 2025).

3. Streaming analytics layer (feature pipelines)

Telemetry is normalised, time aligned, and transformed into features for prediction, detection, and scheduling. The pipeline supports both streaming and batch learning, enabling online evaluation and periodic retraining. High velocity and heterogeneous signals motivate stream processing with clear latency budgets (Aramide, 2025).

4. Learning and decision layer (predict, detect, optimise)

Multiple models are deployed for distinct tasks: congestion prediction, anomaly detection, failure risk forecasting, and workload placement. This layer supports hybrid decision logic that combines predictive scores with constraints and policies appropriate for safety critical industrial settings (Aramide, 2025; Polu, 2024; Zhang et al., 2025).

5. Execution layer (closed loop orchestration)

Decisions trigger concrete actions: reroute flows, adjust queue weights, migrate workloads, scale replicas, or initiate recovery playbooks. The goal is a closed loop system that reduces reaction time and improves resilience relative to controller centric SDN approaches. (Zhang, 2025; Veeravalli, 2026)

6.2 Testbed Strategy and Workload Models

A two-track evaluation strategy will be used to ensure both realism and control.

Track A: Reproducible emulation and simulation

- Network digital twin style evaluation using representative topologies (leaf spine and factory edge clusters) with controllable traffic generators representing distributed training, inference pipelines, and sensor driven bursts (Polu, 2024).
- Supports systematic ablations across topology, load, and failure scenarios.

Track B: Prototype implementation

- A small-scale prototype integrating programmable switching and telemetry collection with an orchestration stack.
- Focus is on validating feasibility of in network telemetry extraction, low latency inference integration, and closed loop control (Aramide, 2025).

Workloads will model:

- **East west traffic** for distributed training and parameter exchange
- **North south traffic** for sensor ingestion, control signals, and monitoring
- **Burstiness** due to machine cycles and production stage synchronisation

These choices align with observed challenges in AI intensive environments where traffic is dynamic and latency sensitive (Aramide, 2025).

6.3 Telemetry Pipeline and Feature Engineering

Telemetry capture

- In band telemetry for per packet or sampled per flow features (Queue Occupancy, Hop Latency Proxies, ECN, Path Identifiers) (Khichane, 2024).
- Host level metrics from compute nodes (CPU and memory utilisation, accelerator utilisation, I/O saturation) (Khichane, 2024).
- Log streams for failure precursors.

Stream processing and storage

- Real time ingestion into a streaming bus, with windowed aggregation (sub second to seconds) for prediction tasks.
- A Lakehouse style store for offline training and retrospective analysis.

Feature engineering

- Temporal features: rolling means, exponential moving averages, rate of change, burst indicators.
- Structural features: topology aware path features and link centrality proxies for congestion prediction.
- Cross layer features: joint representations combining network and compute signals, improving diagnosis of whether congestion originates from routing, scheduling, or host bottlenecks.

Advanced telemetry architectures highlight that high resolution signals are required for effective optimisation and early anomaly detection (Aramide, 2025).

6.4 In Network Learning and Congestion Prediction

A key methodological contribution is shifting time critical inference closer to the data plane to avoid delays caused by repeated control plane communication. Traditional SDN patterns can add latency that is unacceptable for congestion mitigation and intrusion response (Zhang, 2025).

Model families

- **Graph Neural Networks (GNNs)** for topology aware congestion prediction, modelling switches and links as graph structures. GNNs can represent spatial dependencies that time series models may miss when congestion propagates across correlated links. (Aramide, 2025)
- **Time series baselines** such as LSTM for congestion forecasting using link level histories, enabling controlled comparison. (Aramide, 2025)

Deployment strategy

- Use quantisation aware training for smaller inference footprints suitable for constrained environments, consistent with approaches that bring ML inference into the data plane. (Zhang, 2025)
- Partition inference: lightweight feature extraction in the switch, heavier inference at the edge controller when needed.
- Confidence based decisioning: fast local actions for high confidence events, escalation to more expensive inference or global optimisation otherwise. (Zhang, 2025)

6. Anomaly Detection and Drift Monitoring

Because labelled failure and congestion events are rare and evolve over time, the system will use unsupervised and drift aware components.

Anomaly detection

- Autoencoders or Isolation Forest (iForest) for detecting outliers in latency distributions, microbursts, packet loss spikes, and buffer instability (Aramide, 2025).

Drift detection

- Continuous monitoring of feature distributions and prediction residuals to detect when models become stale under changing factory workloads and network conditions. In network drift detection is a key mechanism for maintaining accuracy in dynamic environments (Zhang, 2025).

Outputs feed the orchestration layer as early warnings, enabling intervention before service level violations or production interruptions.

6.6 Predictive Failure Analysis and Self-Healing Control Loops

Predictive failure analysis will be implemented as a multi signal risk scoring pipeline combining telemetry and logs. Traditional threshold monitoring is prone to false positives and does not capture complex precursors (Polu, 2024).

Failure prediction models

- LSTM and Transformer based models for temporal dependencies in logs and metrics.
- Optional graph-based models for correlated failures across network and compute components where failure propagation follows dependency edges (Polu, 2024).

Self-healing actions

- Workload migration or replica scaling when risk exceeds a threshold.

- Traffic steering away from suspected failing links or switches.
- Automated playbooks aligned with AIOps style remediation but triggered by predictive signals rather than reactive alerts (Polu, 2024; Veeravalli, 2026).

This operationalises the concept of self-healing distributed AI infrastructure with closed loop sensing, diagnosis, planning, and execution (Veeravalli, 2026).

6.7 Energy Aware Workload Orchestration

Manufacturing settings impose stricter constraints on power budgets and cooling than typical datacentres. The orchestration policy must therefore jointly optimise latency, reliability, and energy.

Scheduling objectives

- Minimise end to end latency for control critical inference.
- Reduce tail latency under bursty conditions.
- Constrain energy use by shifting non-urgent workloads, consolidating jobs, or selecting energy efficient nodes.

Optimisation methods

- Deep reinforcement learning based scheduling to adapt under uncertainty in task arrivals and device availability, consistent with recent work on industrial edge load balancing under variable offloading (Zhang et al., 2025).
- Constrained optimisation layer to enforce safety constraints and hard production requirements.

6.8 Evaluation Design, Baselines, and Metrics

Baselines

- Controller centric SDN telemetry with centralised decisioning (Zhang, 2025).
- Heuristic routing and ECMP style load balancing.
- Reactive monitoring approaches comparable to AIOps style alerting without predictive risk scoring (Polu, 2024).

Metrics

- Network: mean and tail latency, throughput, packet loss, queue occupancy stability, congestion event frequency.
- Learning: AUC / F1 for failure prediction, precision recall for anomaly detection, calibration error, drift detection lead time.
- System: time to detect and time to mitigate, recovery success rate, production disruption minutes avoided.
- Energy: energy per inference, energy per completed workload, peak power reduction.

Ablations

- With vs without in network inference.
- GNN vs time series predictors.
- With vs without drift detection.

- Predictive remediation vs reactive remediation.

7.0 Research Contributions and Innovation

This research aims to advance the integration of datacentre-grade computing infrastructure with AI-driven manufacturing environments by introducing a network architecture that embeds intelligence directly within the data-plane. The contributions extend beyond existing Software Defined Networking (SDN), AIOps monitoring frameworks, and conventional edge-orchestration approaches.

7.1 AI-Integrated Programmable Network Telemetry

Existing datacentre monitoring systems rely primarily on controller-level telemetry analysis. In traditional SDN architectures, telemetry data must travel from switches to the control plane for analysis before corrective actions are applied. This introduces additional latency and limits the responsiveness required for industrial AI workloads (Zhang, 2025).

The proposed system introduces:

- **In-band network telemetry embedded in programmable switches**
- **Real-time ML inference within the network data plane**
- **Closed-loop telemetry feedback for automated routing optimisation**

Programmable networking technologies such as P4 and eBPF enable packet-level visibility and dynamic flow control within high-performance interconnects (Aramide, 2024).

Compared to existing approaches:

Approach	Limitation	Proposed Contribution
Traditional SDN telemetry	Control-plane dependency introduces delay	Data-plane embedded inference enables microsecond-level responses
Passive monitoring frameworks	Detect anomalies after congestion occurs	Predictive telemetry anticipates congestion
Static routing policies	Cannot adapt to AI workload bursts	Reinforcement learning driven dynamic routing

This architecture transforms network infrastructure from a passive transport layer into an active, self-optimising computational system.

7.2 Predictive Network Optimisation using Graph-Based AI Models

Modern AI workloads generate highly dynamic traffic patterns across datacentre fabrics. Existing traffic engineering approaches rely on heuristics or static policies which are insufficient for complex AI pipelines.

*This research introduces a **multi-model learning framework** capable of performing predictive network optimisation.*

The architecture integrates several complementary models:

Congestion Prediction

Graph Neural Networks (GNNs) model network topology as a graph where:

- nodes represent switches or edge servers
- edges represent network links

GNNs can therefore predict congestion hotspots by learning structural relationships across network paths. This approach improves routing decisions compared with conventional time-series methods.

Anomaly Detection

Unsupervised anomaly detection models will identify abnormal flow patterns caused by:

- packet loss
- latency spikes
- microbursts
- switch buffer overflows

Autoencoders or Isolation Forest models can identify such anomalies without labelled data (Aramide, 2025).

Adaptive Traffic Control

Reinforcement learning agents dynamically adjust:

- flow routing
- bandwidth allocation
- queue management policies

Deep Q-Networks can learn policies that minimise network congestion and latency under variable workloads.

Compared with existing congestion management techniques:

Method	Limitation	Improvement Proposed
ECMP / static load balancing	unaware of traffic dynamics	topology-aware AI routing
heuristic congestion control	reactive rather than predictive	proactive congestion forecasting
controller-based ML models	latency from data-plane communication	embedded inference in network hardware

7.3 Predictive Failure Detection and Self-Healing Infrastructure

Industrial AI systems require extremely high reliability. Failures in network nodes, servers, or edge devices can interrupt production processes.

Existing monitoring frameworks detect failures only after they occur.

This research proposes a predictive failure analysis framework using AI models trained on system telemetry.

Deep learning models such as:

- Long Short-Term Memory networks

- Transformer architectures
- Graph Neural Networks

have demonstrated superior performance in learning temporal failure patterns from system logs and telemetry streams (Polu, 2024).

These models will analyse:

- CPU utilisation patterns
- network latency trends
- system log events
- hardware health metrics

The predictive system will generate failure probability scores that trigger automated recovery mechanisms including:

- workload migration
- dynamic routing adjustments
- resource reallocation

This approach aligns with emerging self-healing infrastructure models in distributed AI systems (Veeravalli, 2026).

7.4 Energy-Aware Workload Scheduling for Industrial AI

Manufacturing environments impose stricter energy constraints than traditional datacentres.

The proposed architecture integrates energy-aware workload orchestration across edge and datacentre nodes.

Deep Reinforcement Learning based schedulers will optimise:

- AI task placement
- Compute resource utilisation
- Network bandwidth allocation

Recent research has shown that DRL-based workload scheduling can reduce system latency while improving resource balancing across distributed nodes (Zhang et al., 2025).

By combining telemetry-driven monitoring with energy-aware scheduling policies, the proposed system can achieve:

- Lower operational energy consumption
- Improved system stability
- Reduced production downtime.

8.0 Research Timeline

Year	Research Activities	Outputs
Year 1	Literature review on AI networking and programmable data planes. Design architecture for AI-driven telemetry framework. Develop programmable telemetry pipeline using P4 and SDN.	Architecture design paper, conference submission
Year 1–2	Develop machine learning models for anomaly detection and congestion prediction. Build simulation environment using Mininet or ns-3. Train models using synthetic AI workloads.	ML model framework and dataset publication
Year 2	Implement in-network learning architecture with embedded inference capabilities. Integrate telemetry pipelines with ML models for real-time optimisation.	Experimental prototype and journal paper
Year 2–3	Design reinforcement learning system for adaptive routing and energy-aware scheduling. Deploy predictive failure detection models using telemetry data.	Journal paper on AI-driven network optimisation
Year 3	Large-scale system evaluation and benchmarking against state-of-the-art methods. Final optimisation and thesis writing.	Thesis submission and final publications

9.0 Conclusion

This research addresses a critical and unresolved challenge in the evolution of AI-driven manufacturing: the inability of existing SDN, AIOps, and edge orchestration paradigms to deliver deterministic, low-latency, and resilient behaviour under industrial-grade AI workloads. By reconceptualising networks as intelligent, autonomous systems rather than passive conduits, and by embedding high-resolution telemetry, in-network machine learning, and self-healing mechanisms directly into the infrastructure, the proposed framework enables real-time adaptation that cannot be achieved through control-plane-driven or post-hoc optimisation approaches. The research therefore advances a principled pathway for merging datacentre-scale performance with manufacturing constraints, supporting resilient, low-latency, and energy-aware industrial AI systems.

References

- Aramide, O.O., 2024. Programmable Data Planes (P4, eBPF) for High-Performance Networking: Architectures and Optimizations for AI/ML Workloads. *SAMRIDDHI: A Journal of Physical Sciences, Engineering and Technology*, 16(02), pp.108-117.
- Aramide, O.O., 2025. Advanced Network Telemetry for AI-Driven Network Optimization in Ultra Ethernet and InfiniBand Interconnects. *SAMRIDDHI: A Journal of Physical Sciences, Engineering and Technology*, 17(01).
- Chinnaraju, A., 2024. Real-Time Adaptive AI pipelines for edge-cloud systems: Dynamic optimization based.
- Gill, S.S., Golec, M., Hu, J., Xu, M., Du, J., Wu, H., Walia, G.K., Murugesan, S.S., Ali, B., Kumar, M. and Ye, K., 2025. Edge AI: A taxonomy, systematic review and future directions. *Cluster Computing*, 28(1), p.18.
- Khichane, A., 2024. Diagnostic of performance by data interpretation for 5G cloud native network functions (Doctoral dissertation, Université Paris-Saclay).
- Manzoor, S., Raza, A. and Domzal, J., 2024, December. Advance networking paradigms: Integrating artificial intelligence into software-defined wireless networks. In *2024 International Conference on Frontiers of Information Technology (FIT)* (pp. 1-6). IEEE.
- Polu, O.R., 2024. AI-driven prognostic failure analysis for autonomous resilience in cloud data centers. *J. ID*, 2563, p.4512.
- Veeravalli, K.S.S., 2026. Self-Healing Distributed Networks For AI Systems: A Paradigm Shift In Resilient Architecture. *Journal of International Crisis and Risk Communication Research*, 9(1), p.173.
- Zhang, F., Jiang, Y., Bao, X., Zhou, X., Zong, Y., Liang, X. and Yang, K., 2025. Adaptive Load Balancing for Industrial Edge Computing Systems: An AxTD3-Deep Reinforcement Learning Approach. *IEEE Transactions on Network Science and Engineering*.
- Zhang, K., 2025. Intelligent Data-Planes for Network Traffic Management (Doctoral dissertation, Université d'Ottawa| University of Ottawa).